

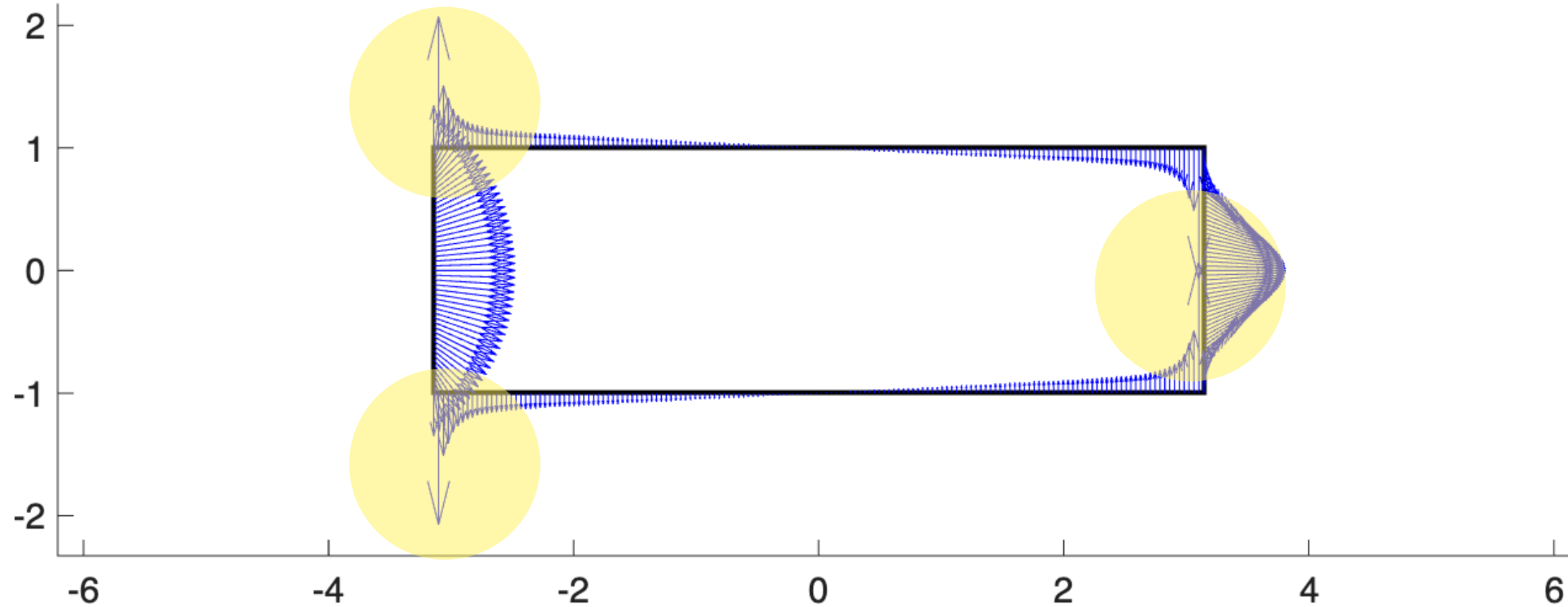
# Segment Method Updates as of April 17th...

- **Segment Method (for both Stokeslets and Source Doublets) is fully coded and debugged for the infinite channel problem!** Results on slides 2-5
  - From previous meeting: There was an issue in the source doublet velocity to gforce file, but that has since been debugged!
  - Codes are set-up so that we can easily switch between blobs ( $\phi$  and  $\psi$ ). See list of function files uploaded to GitHub on slide 10.
- **Implemented Segment Method 2 ways**
  - $\beta$  constant on each segment: On each segment, use the average of beta as the constant beta value:  $\beta_0 = (\beta_{start} + \beta_{end})/2$ . See overleaf for details on the formulation.
  - $\beta$  linear on each segment:  $\beta(\alpha) = \beta_{start} + \alpha(\beta_{start} - \beta_{end})$ . I did not write up the formulation in overleaf, as this did not produce any improvement in the method. (Slides 6-7)
- **Continuity where we transition from known (solid) boundary velocity to unknown (perm) boundary velocity remains an issue!** Included possible options for addressing in slides 8-9.

# Segment Method Applied to Permeable Channel

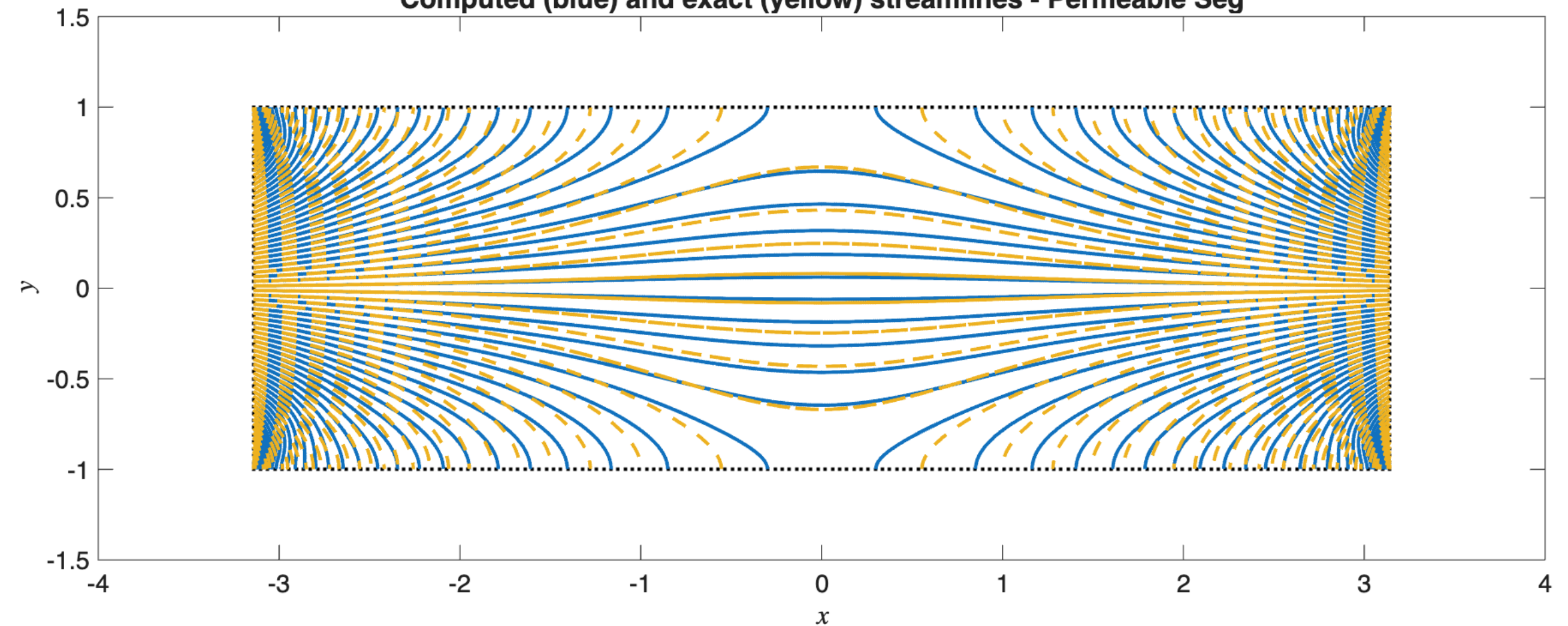
Continuity at the corners (transitioning from known (solid) boundary velocity to unknown (perm) boundary velocity) remains an issue. This is coming from the discontinuity in our doublet forces (g).

Recovered Bd Velocities on Boundaries using segment method

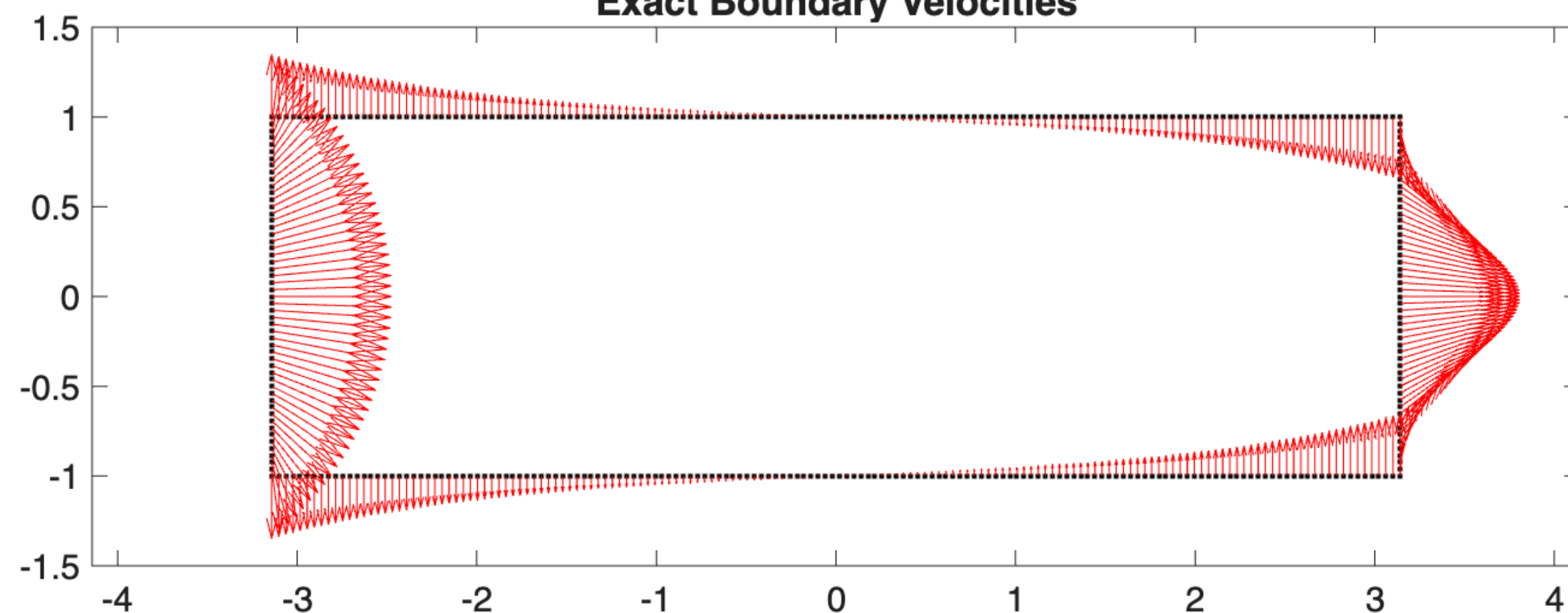


Set-up:  $C = 0.13$  ( $\beta = CDa\epsilon$ ),  $ep = ds\_y/10$ , blob  $\phi$

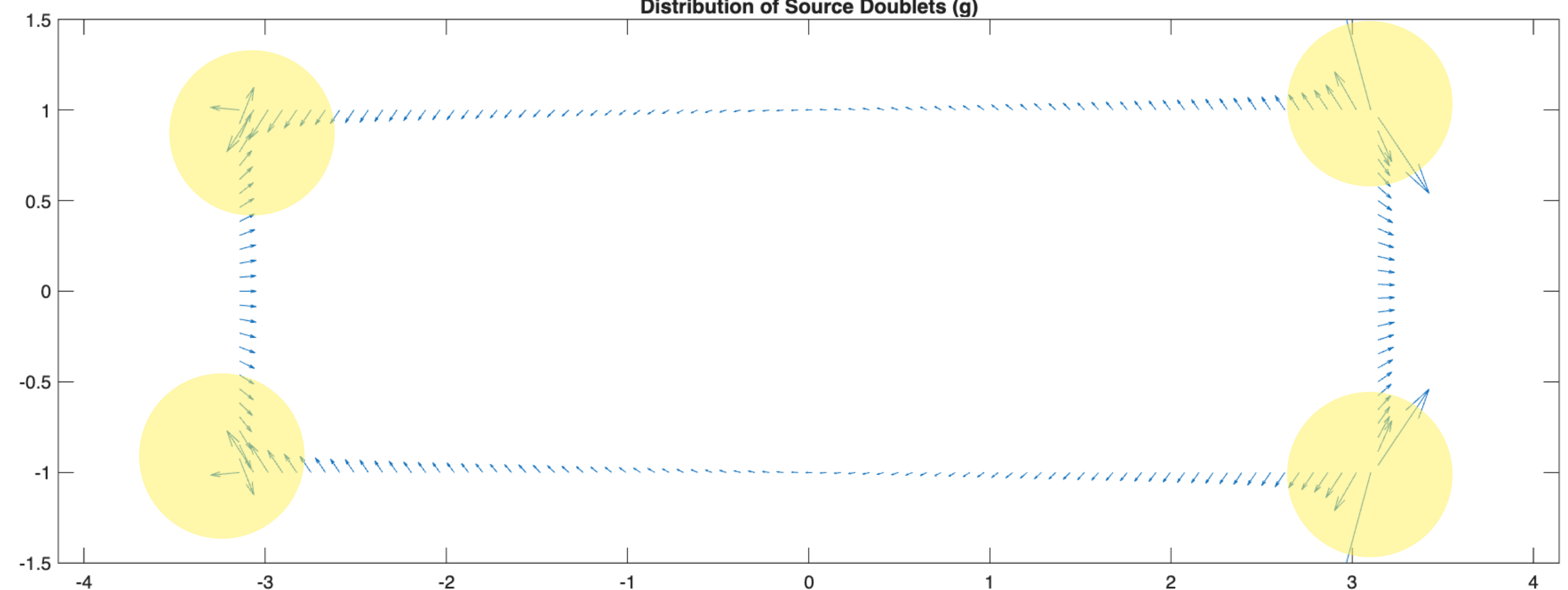
Computed (blue) and exact (yellow) streamlines - Permeable Seg



Exact Boundary Velocities



Distribution of Source Doublets (g)



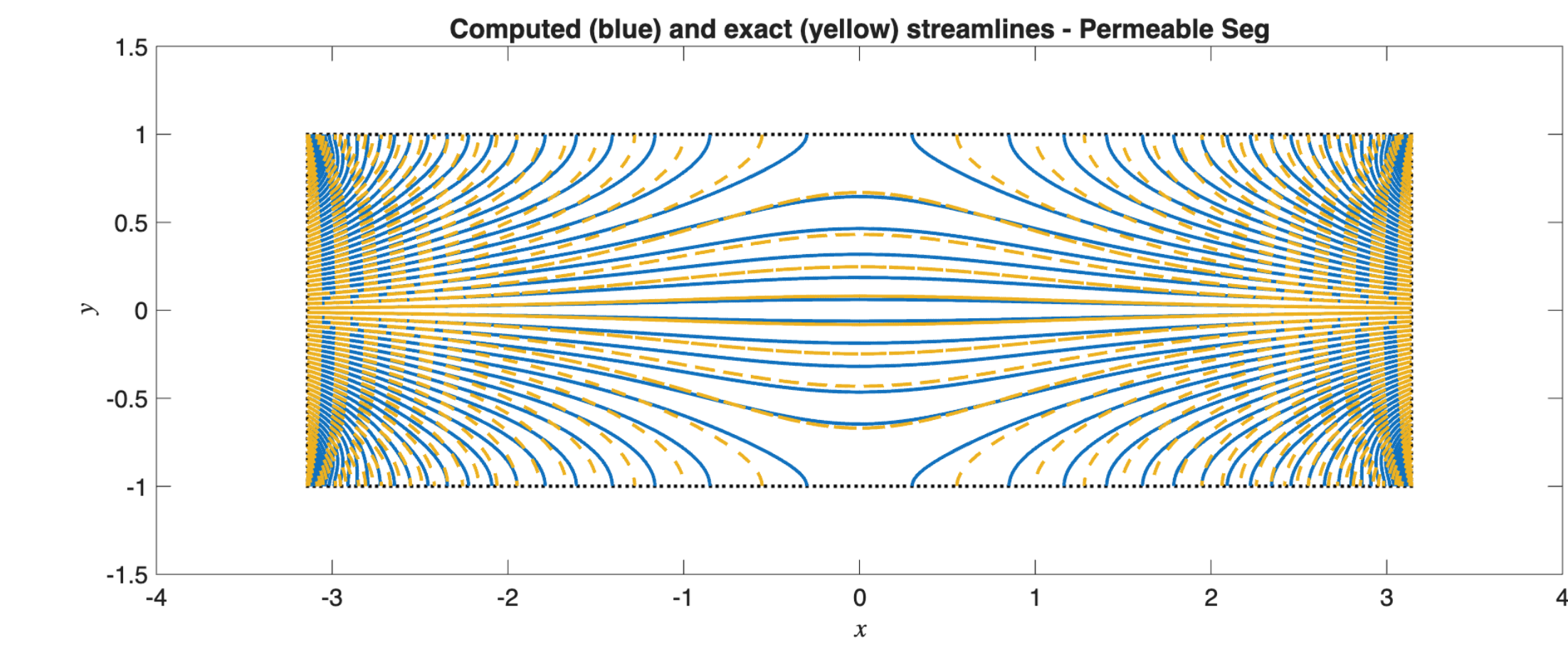


# Permeable Segments - Refining N, blob $\phi$

The velocity discontinuity at the corners gets worse as we refine!

Set-up:  $C = 0.13$  ,  $ep = ds\_y$ , blob  $\phi$

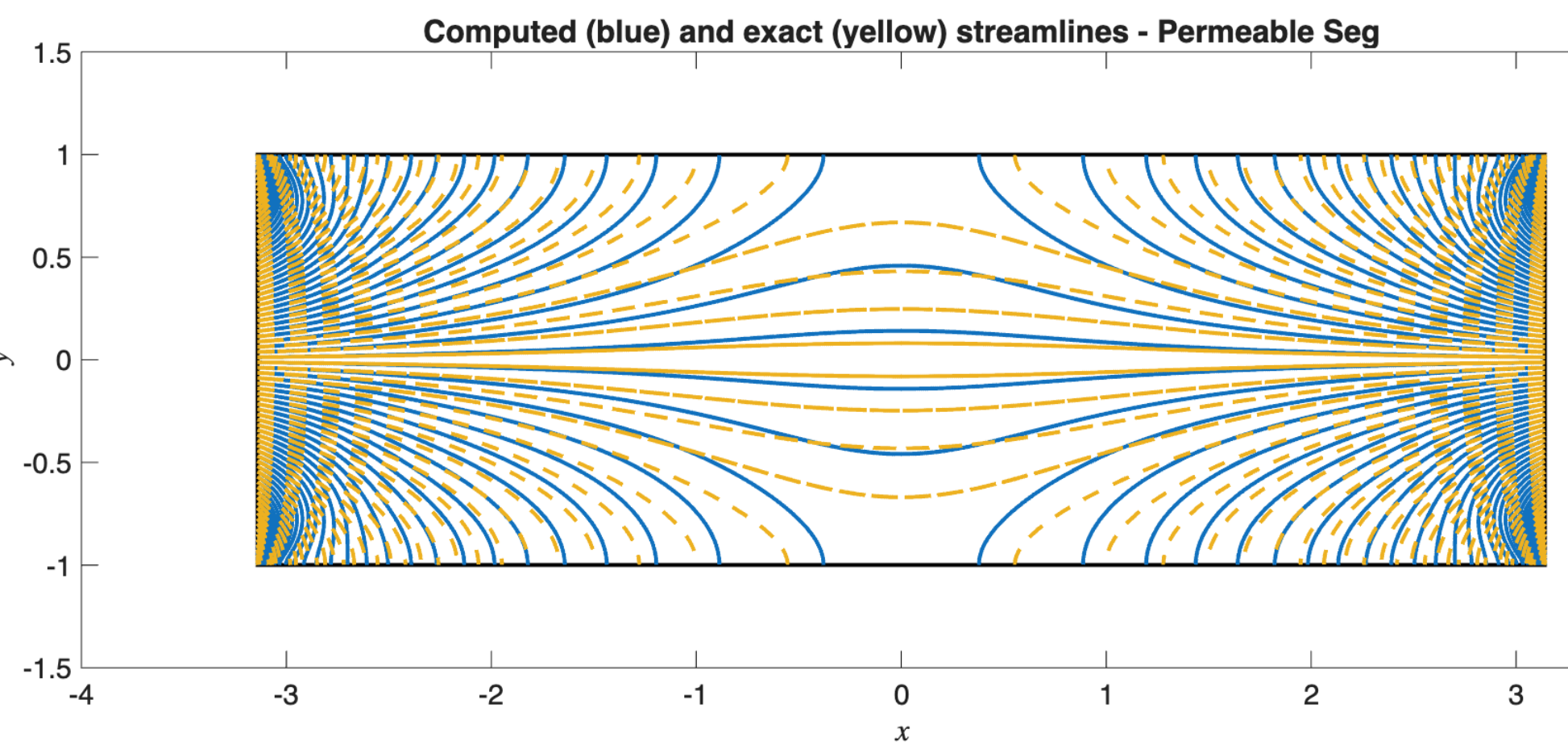
$Ny1 = 640$



$Ny1 = 160$

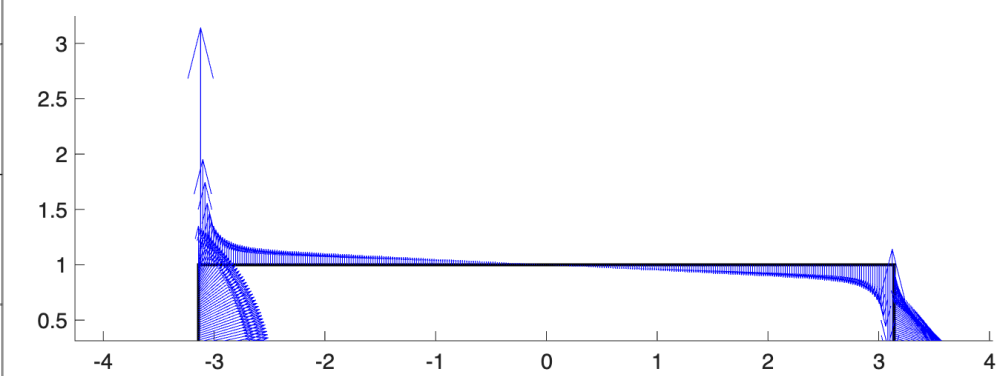
| Max    | L1     | L2     |
|--------|--------|--------|
| 4.7232 | 9.8061 | 2.7447 |
| 12.972 | 6.0376 | 3.0245 |

Recovered Bd Velocities on Boundaries using segment method



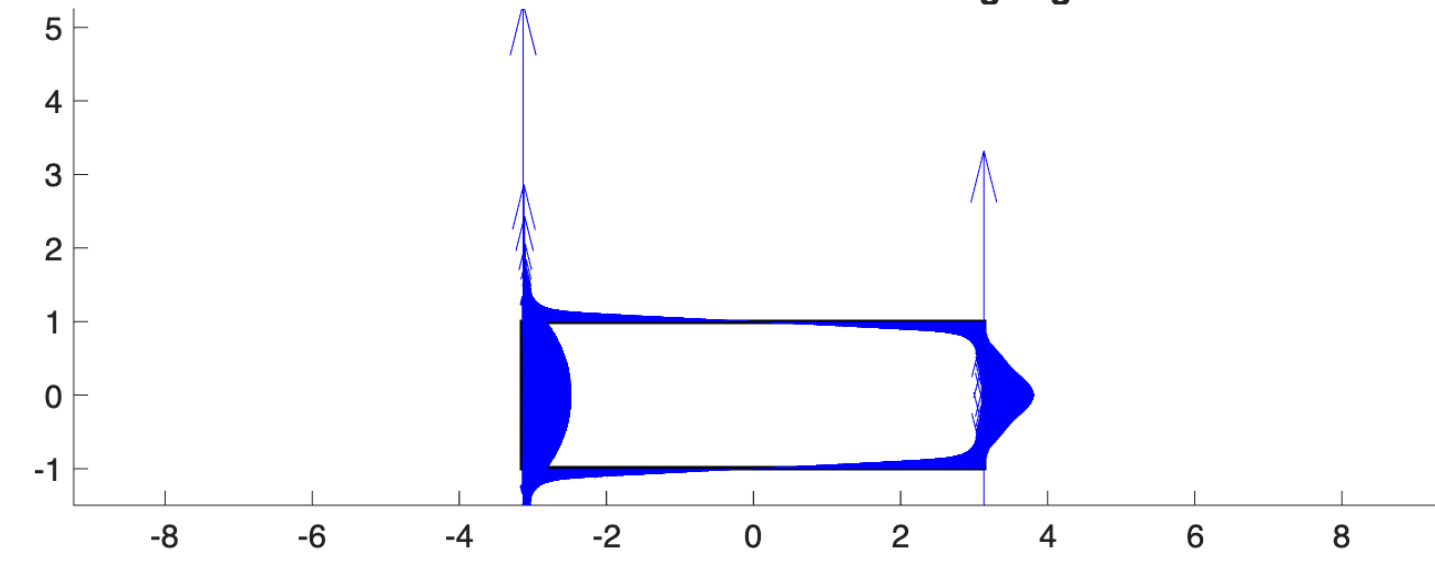
$Ny1 = 320$

| Max    | L1     | L2     |
|--------|--------|--------|
| 10.338 | 1.926  | 1.8329 |
| 24.007 | 1.7937 | 2.2275 |



| Max    | L1     | L2     |
|--------|--------|--------|
| 20.488 | 1.2624 | 1.5421 |
| 27.598 | 0.544  | 1.4158 |

Recovered Bd Velocities on Boundaries using segment method



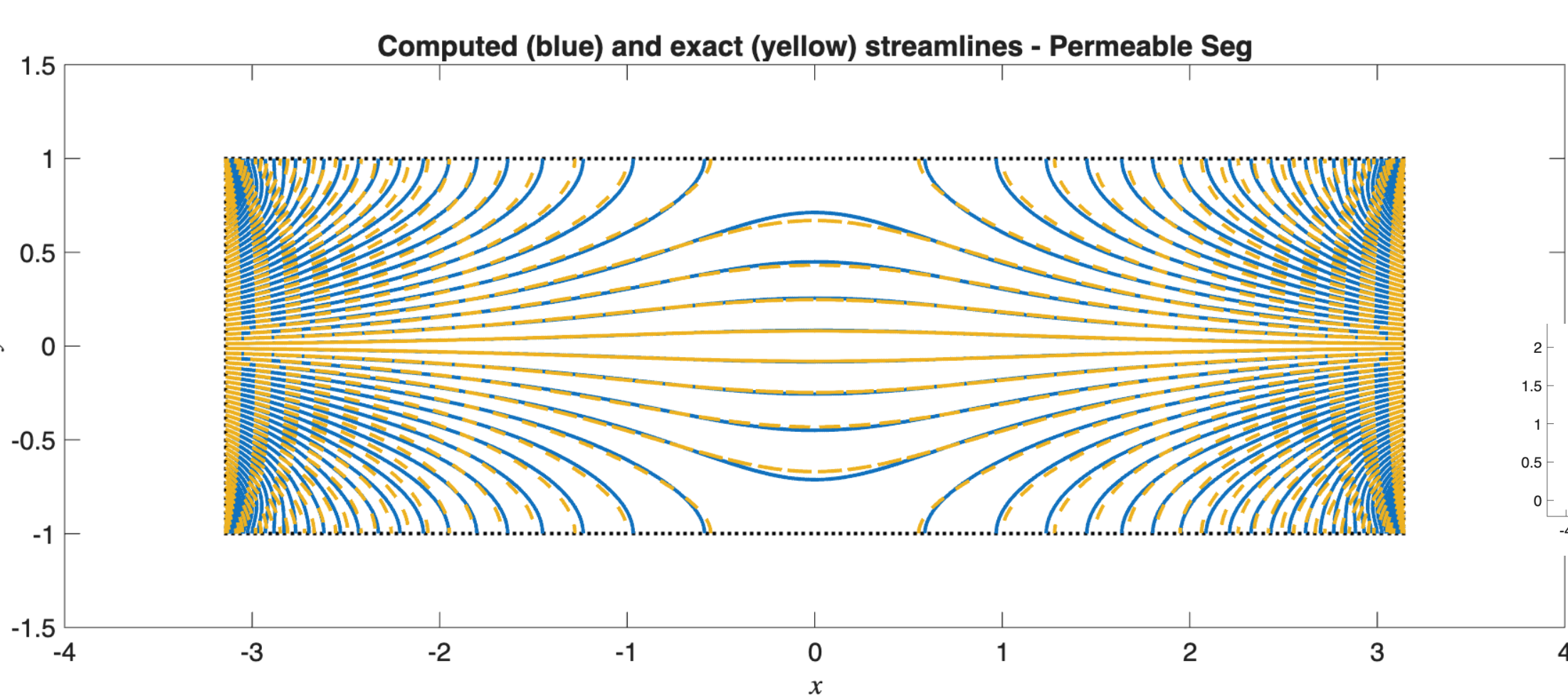


# Permeable Segments - Refining N, blob $\psi$

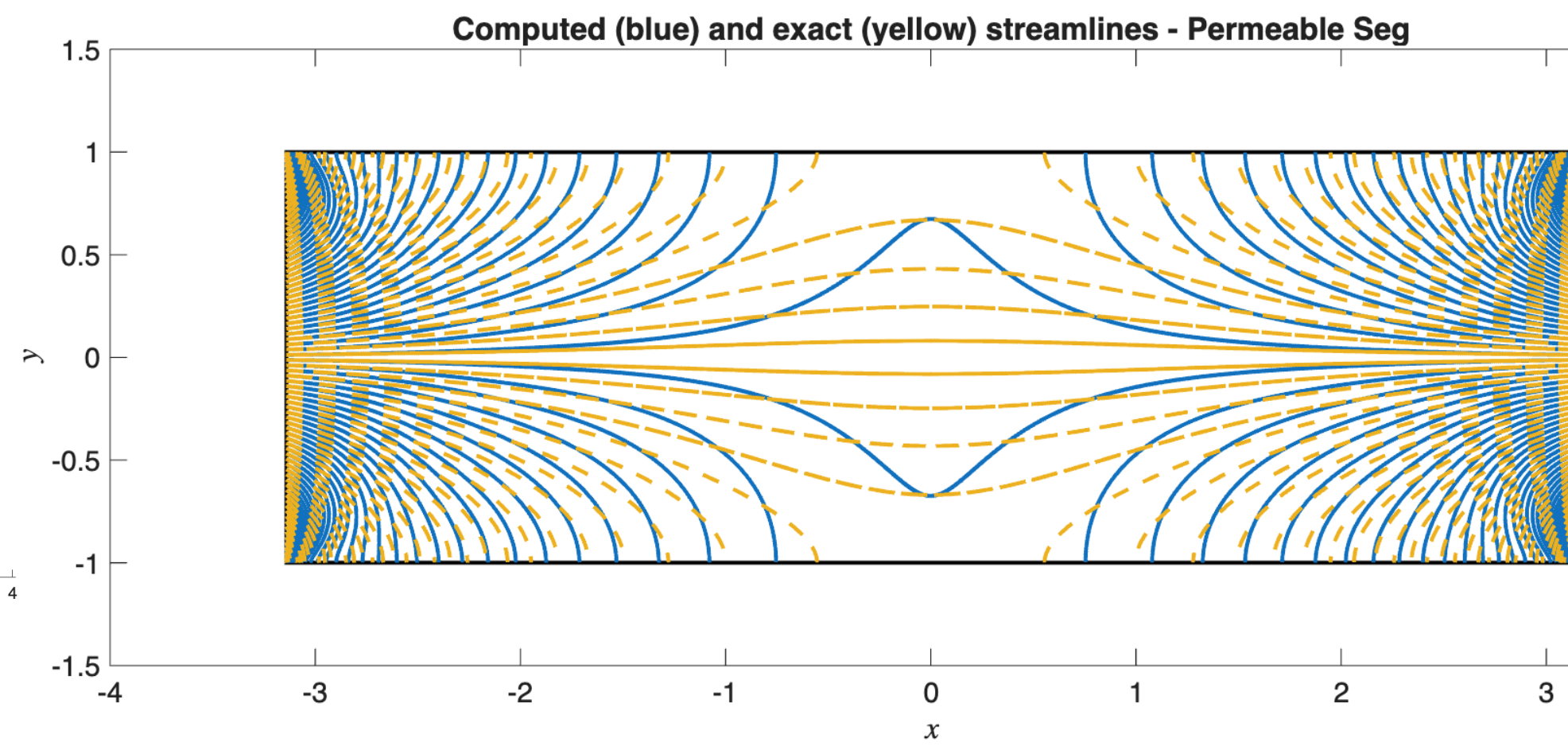
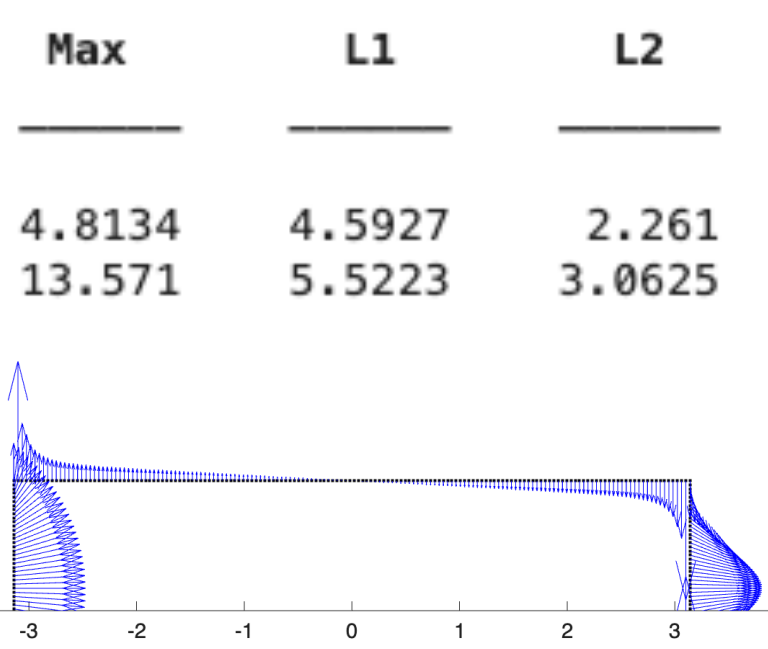
The velocity discontinuity at the corners gets worse as we refine!

Set-up:  $C = 0.053$  ,  $ep = ds\_y$ , blob  $\psi$

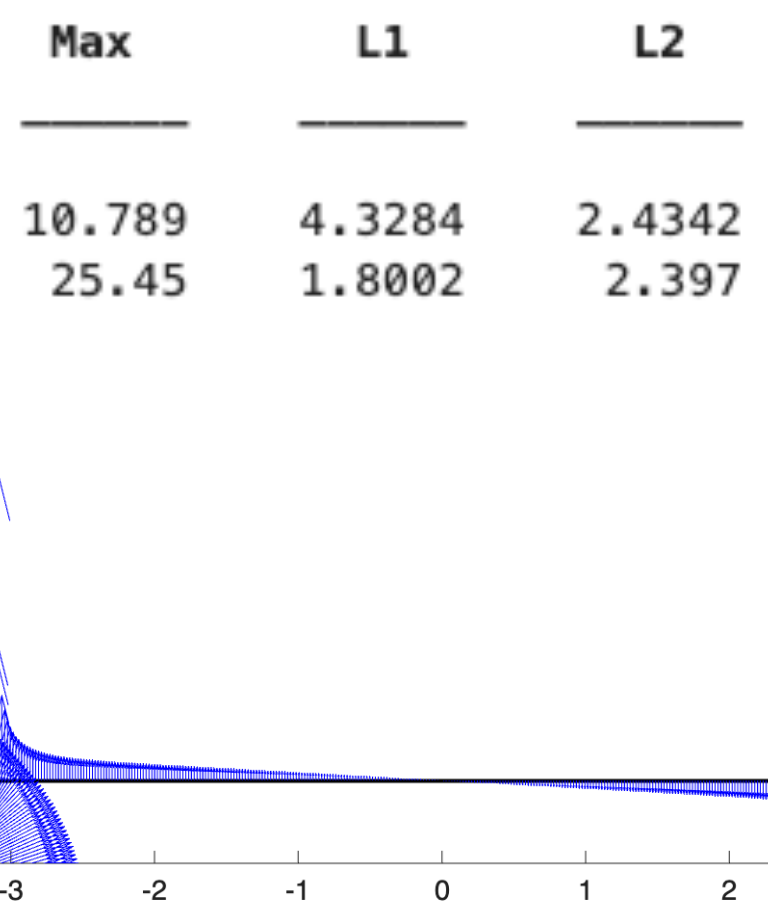
$Ny1 = 640$



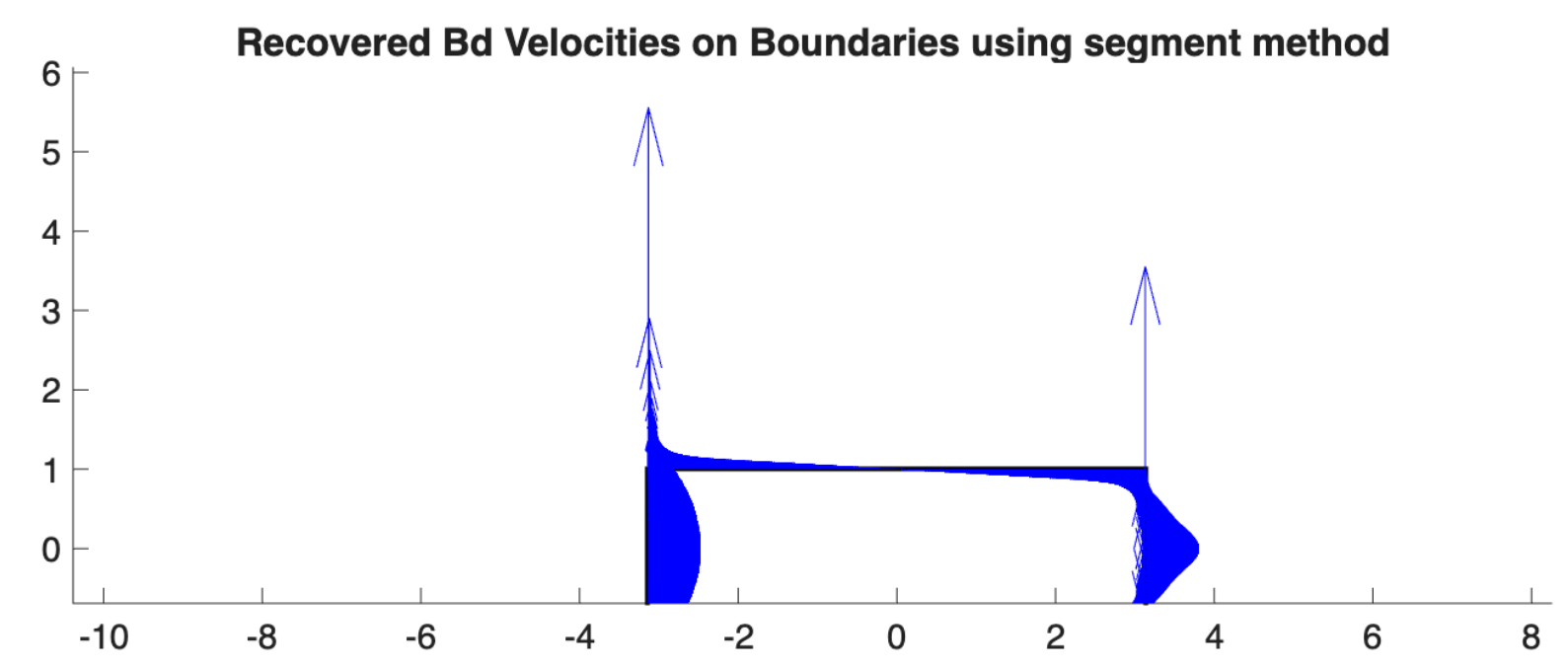
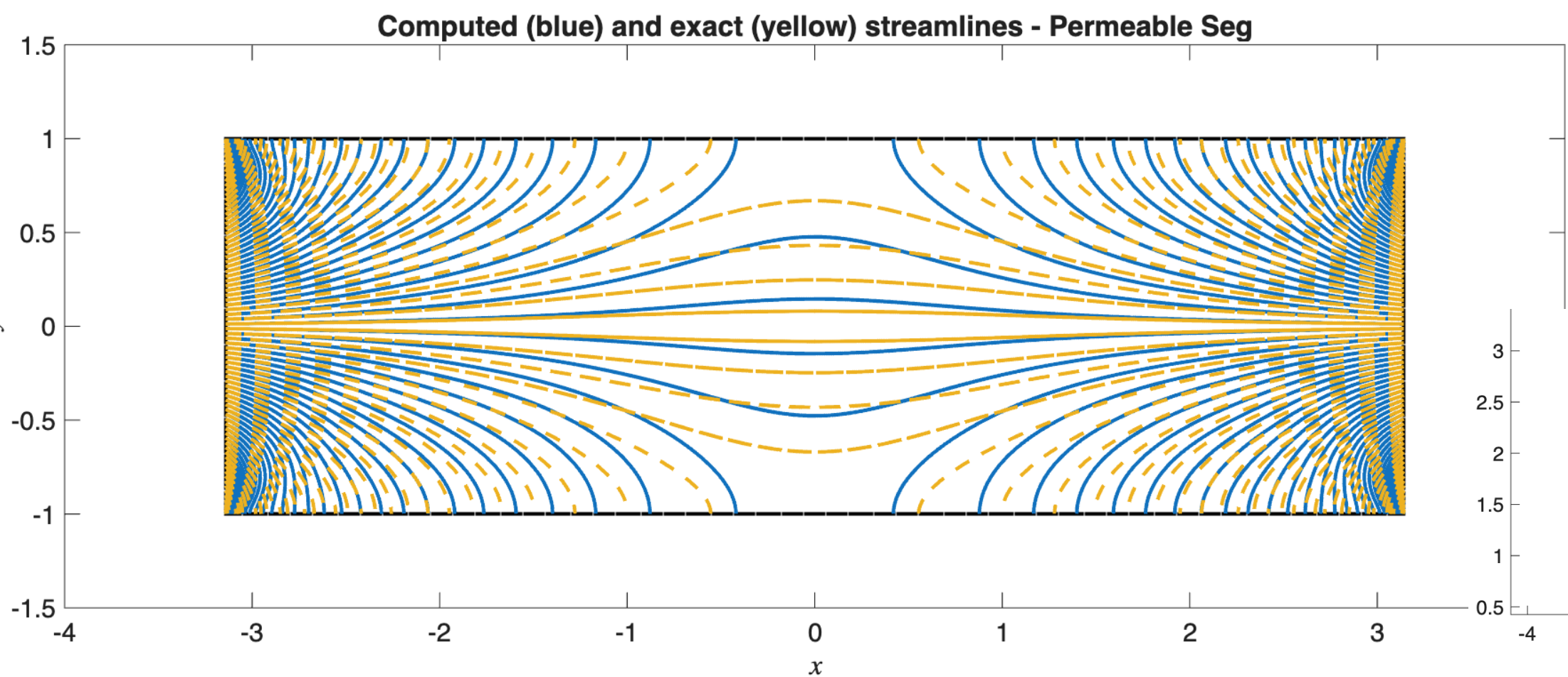
$Ny1 = 160$



$Ny1 = 320$



| Max    | L1      | L2     |
|--------|---------|--------|
| 21.298 | 1.9149  | 1.9892 |
| 29.051 | 0.56938 | 1.5379 |

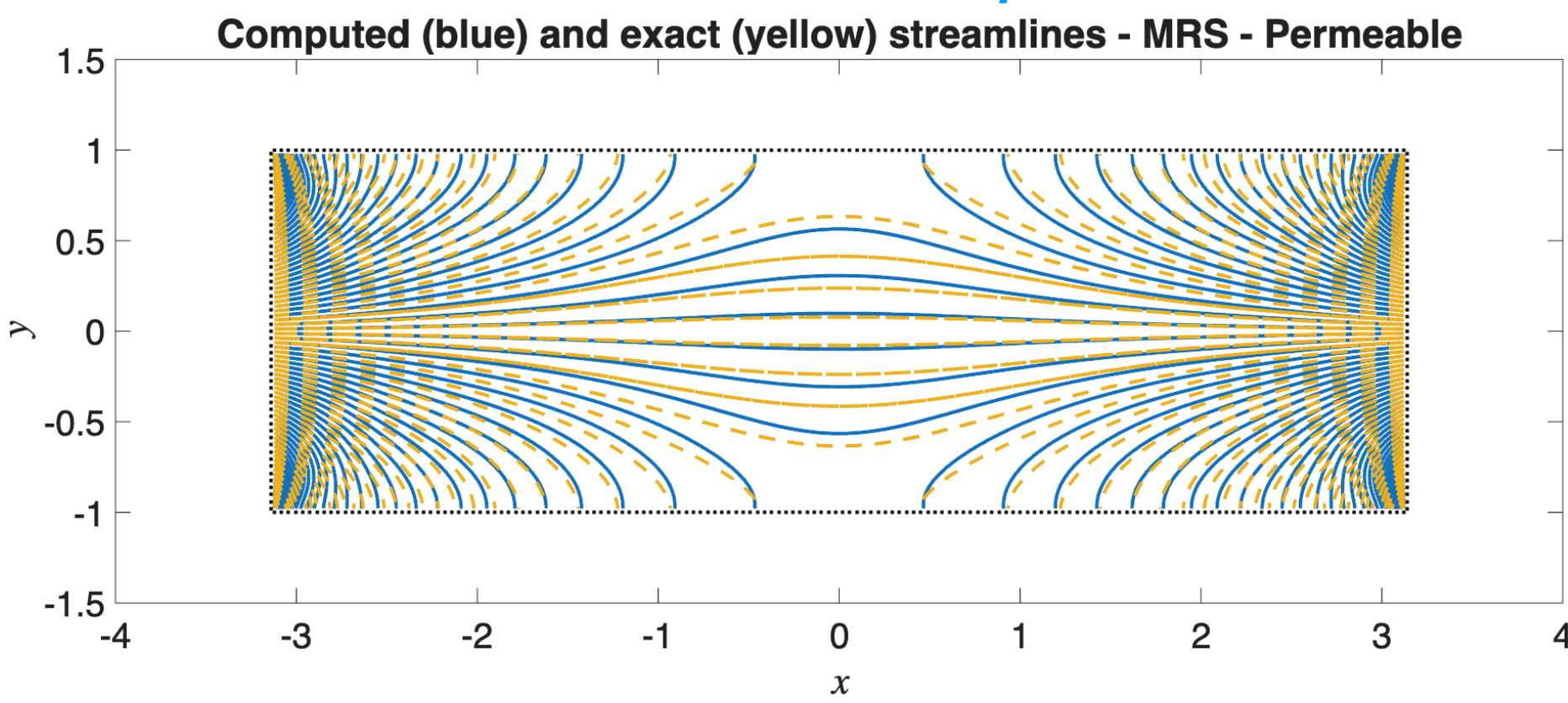




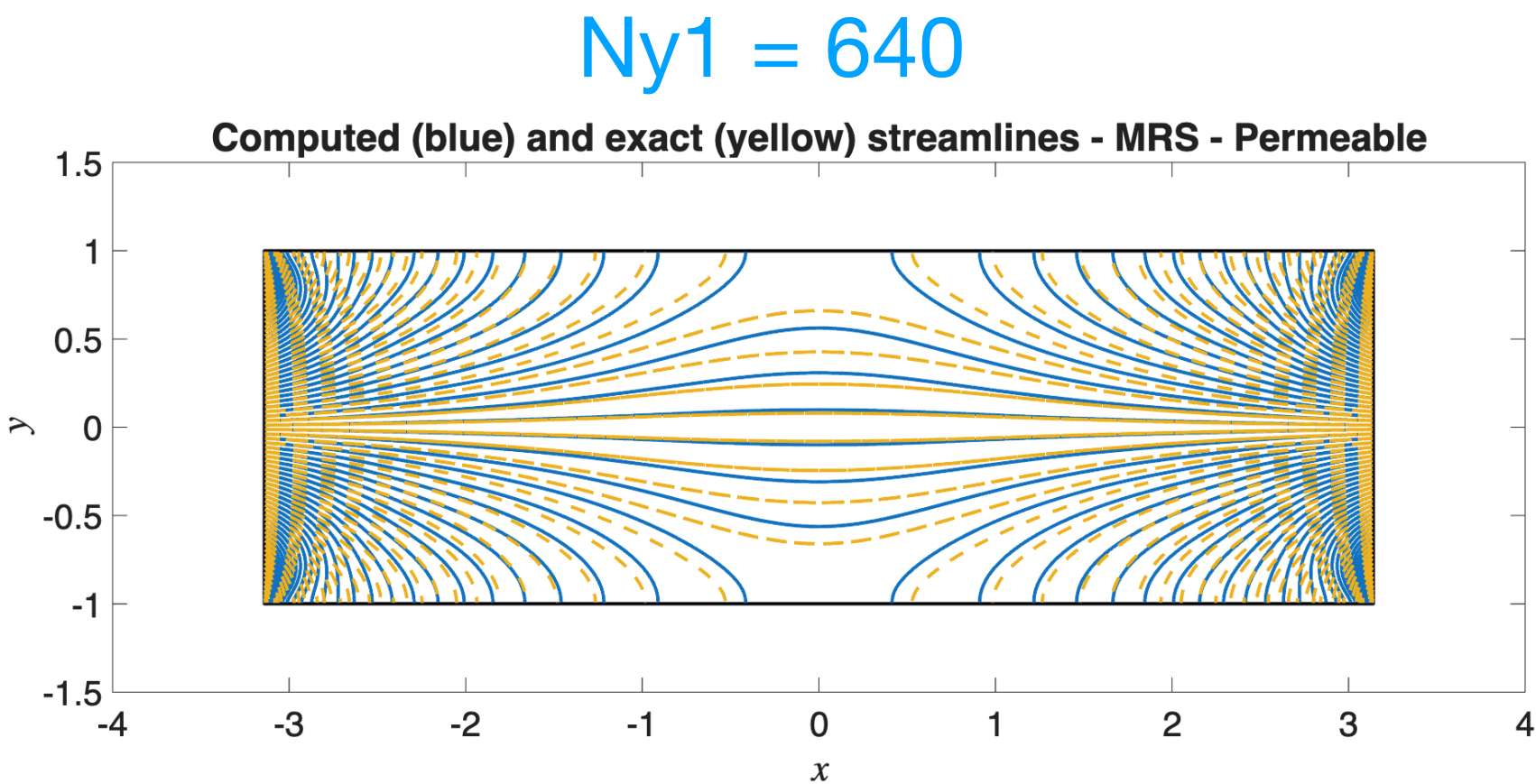
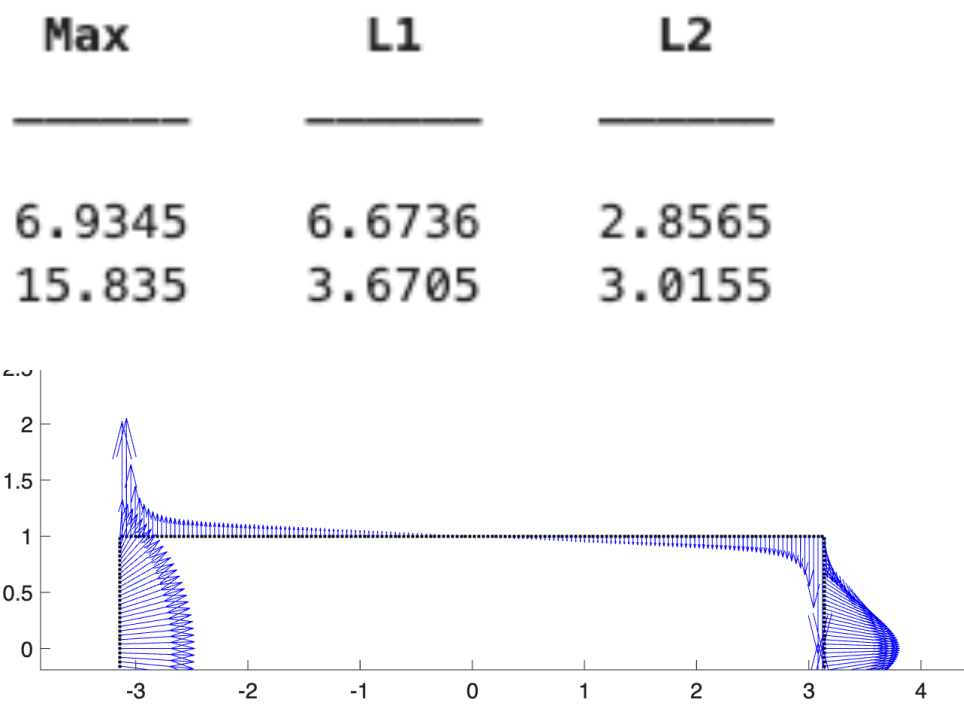
# MRS Comparison - Refining N

Similar to segment method, the velocity discontinuity at the corners gets worse as we refine!

$C = 0.12$  ,  $ep = ds\_y$ , blob  $\phi$ , Step 1 is Stokeslets velocity-to-force



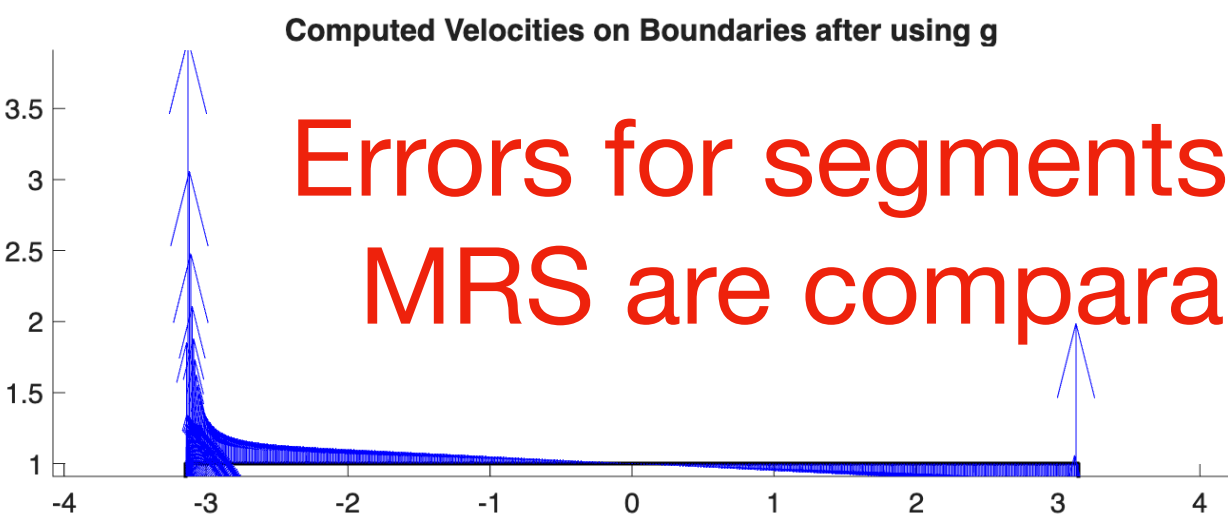
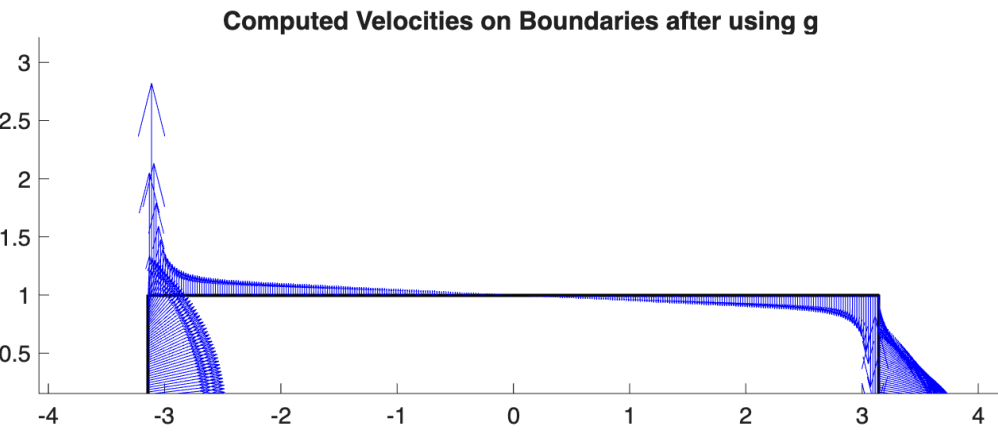
$Ny1 = 160$



| Max    | L1     | L2     |
|--------|--------|--------|
| 15.692 | 7.2515 | 3.6622 |
| 24.395 | 4.3842 | 3.8085 |

$Ny1 = 320$

| Max    | L1     | L2     |
|--------|--------|--------|
| 10.011 | 6.2427 | 3.1416 |
| 22.15  | 4.0546 | 3.4893 |



Errors for segments and MRS are comparable

# Experimenting with $\beta$ in segment method

- **Current treatment:** We have a  $\beta$  value at each node ( $\beta_j$  at node  $y_j$ ). In the segment method, we use a constant  $\beta$  value,  $\beta_0$ , so that on the segment connecting nodes  $y_j$  to  $y_k$ , we have  $\beta_0 = (\beta_j + \beta_k)/2$  (average of  $\beta$  at the segment endpoints). We then assume that  $\mathbf{g}$  varies linearly on a segment. Thus on a segment we have

$$\beta \mathbf{g}(\alpha) = \beta_0(\mathbf{g}_j + \alpha(\mathbf{g}_k - \mathbf{g}_j))$$

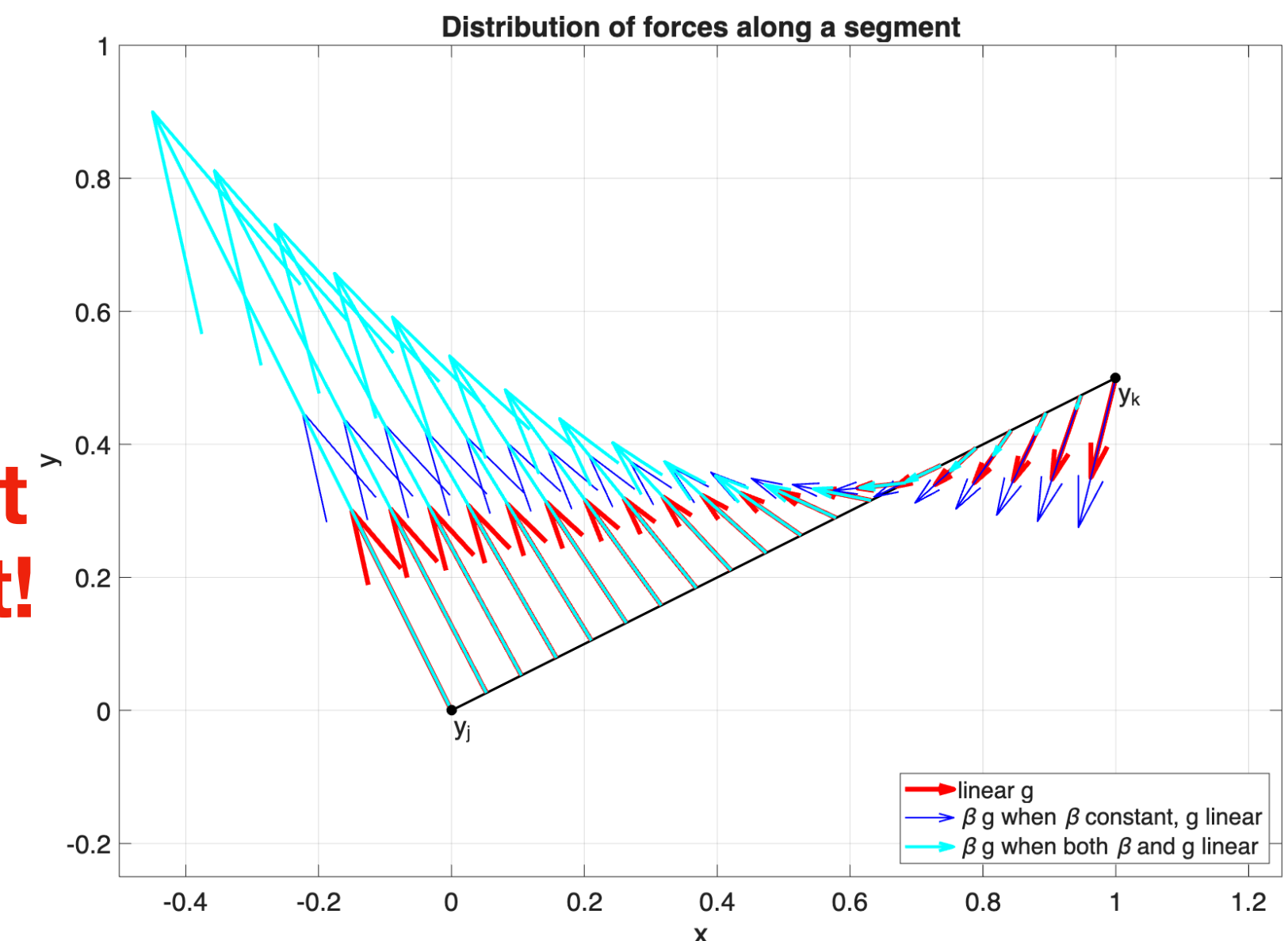
- **Alternate treatment:** Suppose that both  $\beta$  and  $\mathbf{g}$  vary linearly on a segment. Then

$$\beta(\alpha)\mathbf{g}(\alpha) = \beta_j\mathbf{g}_j + \alpha \left( \beta_j(\mathbf{g}_k - \mathbf{g}_j) + (\beta_k - \beta_j)\mathbf{g}_j \right) + \alpha^2(\beta_k - \beta_j)(\mathbf{g}_k - \mathbf{g}_j)$$

- This changed the formulation, since we will end up with higher powers of  $\alpha$  and more integrals of form

$$S_{1,2}^n = \int_0^1 \alpha^n S_{1,2}(r) d\alpha.$$

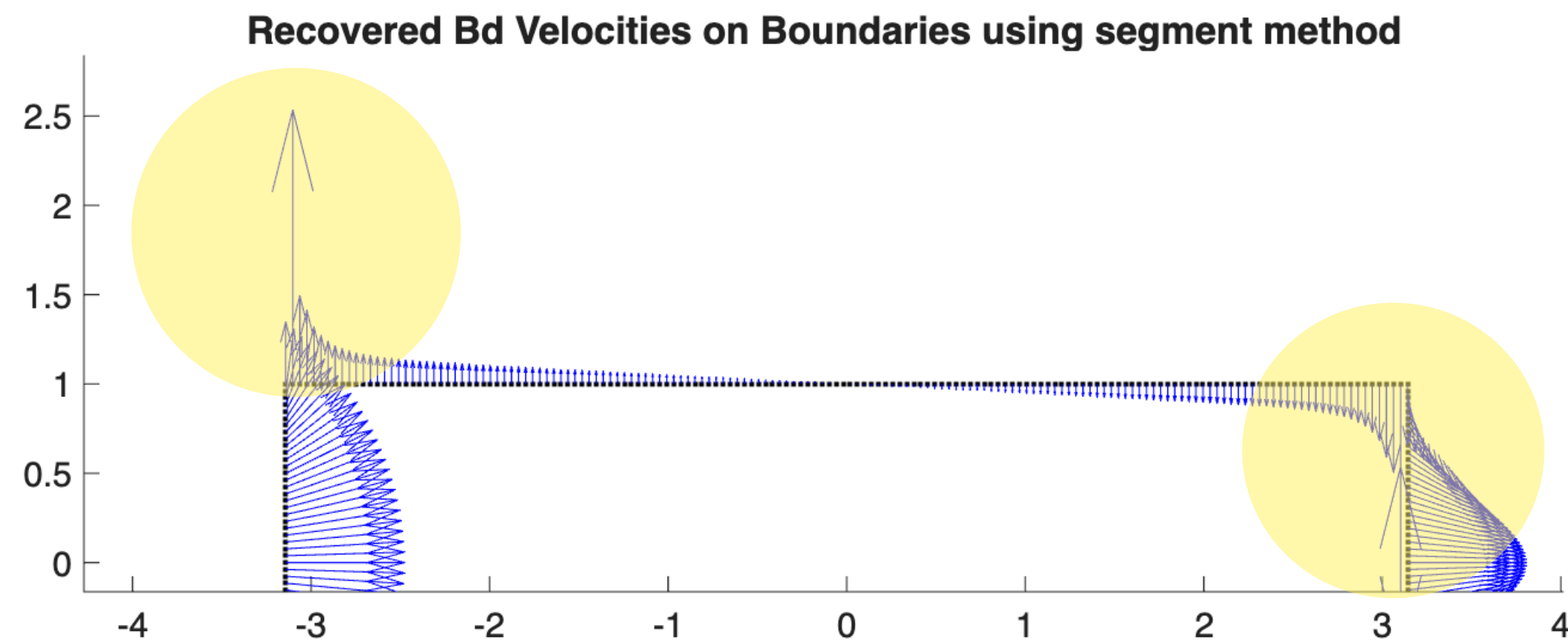
- **Gave this a try to see if it improved the continuity issues at the corners, but unfortunately didn't see any improvement! (see next slide)**





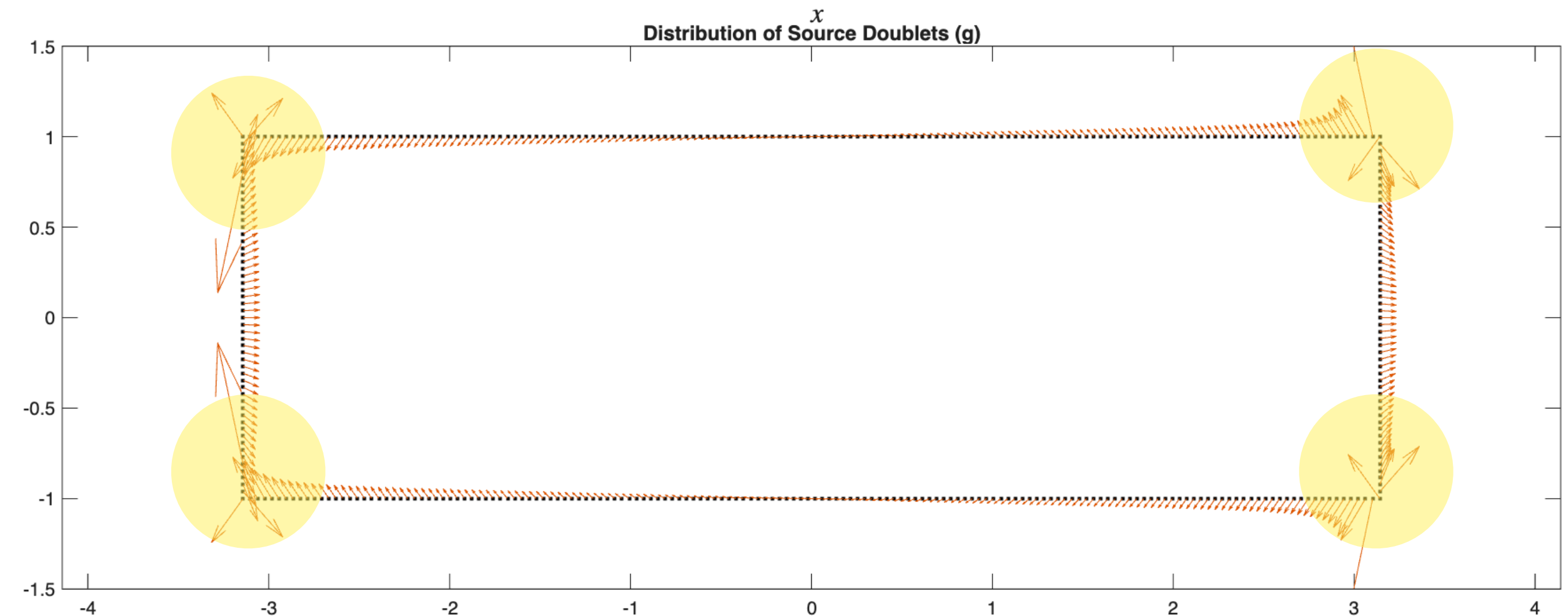
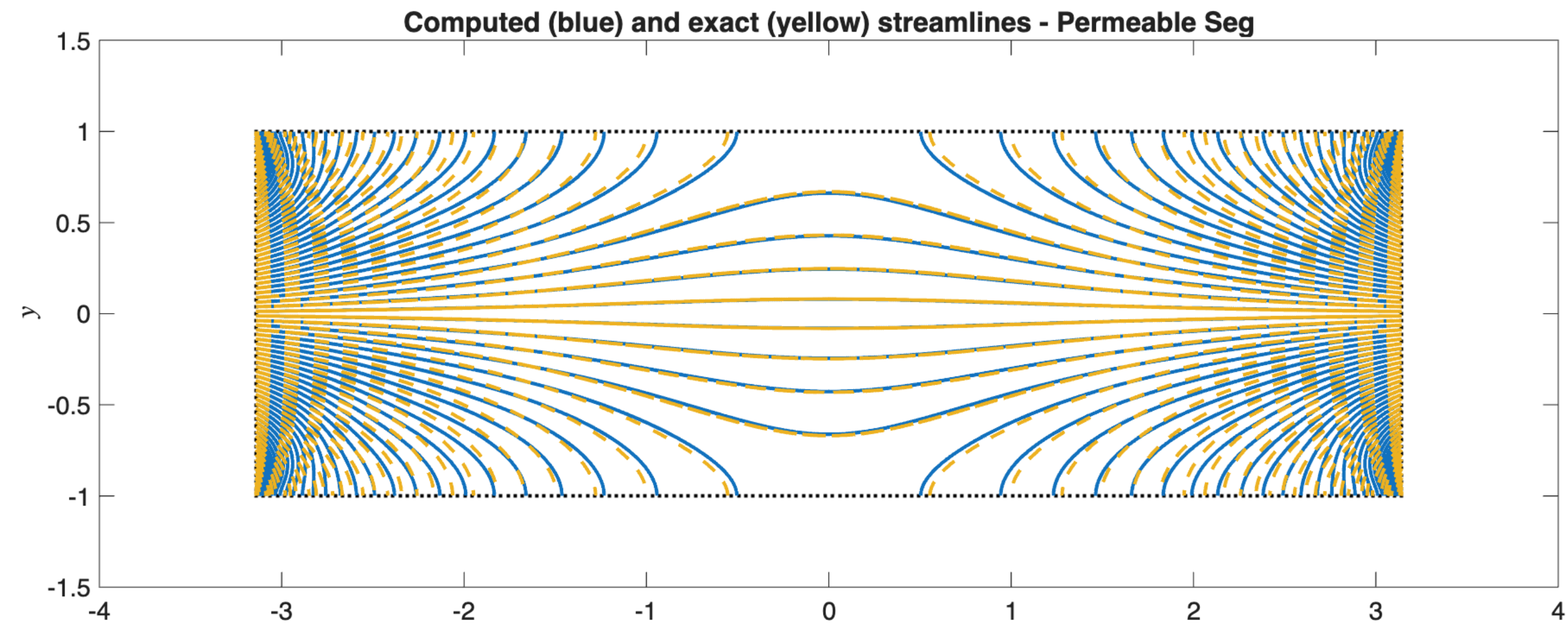
# Segment Method- $\beta$ treated as linear on each segment

Set-up:  $C = 0.0496$  ( $\beta = CDa\epsilon$ ),  $ep = ds\_y/10$ , blob  $\psi$



This did not significantly impact the continuity issues at the corners! If anything, there is a bigger jump at first points in the interior of the permeable region.

| Max    | L1     | L2     |
|--------|--------|--------|
| 7.9175 | 6.1736 | 3.0941 |
| 21.258 | 6.7601 | 4.1319 |



# Enforcing Continuity at the corners

- **Strong Continuity Ideas**

- Tried enforcing  $M_{SD}(\mathbf{y}_{corner}, \mathbf{y})\mathbf{g} = u_{solid}(\mathbf{y}_{corner})$  in Step 1, taking  $\mathbf{y}_{corner}$  out of  $(M_{st} + M_{SD})\mathbf{g} = u_{solid}$  for points in the solid region while still using  $M_{st}\mathbf{g} = \mathbf{0}$  for points in the permeable section. This made the discontinuity worse.
- Other ideas?
- **Some other kind of compatibility condition between the Stokeslet and Source Doublet matrices?** In my head, I thought that treating  $\beta$  as linear on each segment was somehow a way of doing this, but are there other things we could try?
- **Weak Continuity Ideas:** Penalty regularization in Step 2 of Channel Process? I tried sketching it out on the next slide.



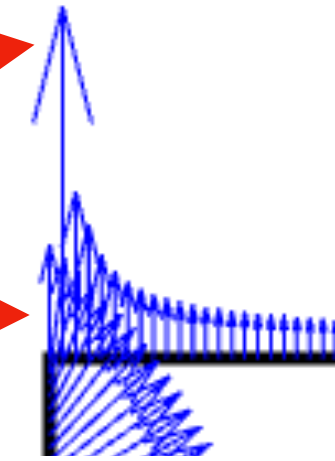
# Idea for Weak Continuity via Penalty Method

(Half-baked idea)

## Proposed Channel Process using a Penalty Regularization in Step 2

1. Solve for  $\mathbf{g}$  on  $\Gamma$  (same as before).  $(\mathcal{M}_{St} + \mathcal{M}_{SD})\mathbf{g} = \mu\mathbf{u}_b$  on  $\Gamma \setminus \Gamma_\beta$ ,  $\mathcal{M}_{St}\mathbf{g} = \mathbf{0}$  on  $\Gamma_\beta$
2. Instead of  $\mathbf{u}_{perm} = M_{SD}\mathbf{g}$ ,  $\mathbf{u}_{perm}$  is the minimum of

$$F(\mathbf{u}_{perm}) = 1/2 \|\mathbf{u}_{perm} - M_{SD}\mathbf{g}\|^2 + \sum_{i \in idxT} \frac{\gamma_i}{2} (u_i^{perm} - u_i^{solid})^2$$



where  $idxT$  is the set of indices for points in the permeable region immediately following transition points (in our case, just inside the corners),  $u_i^{solid}$  is the known boundary velocity at the actual transition point, and  $\gamma_i$  is our penalty parameter. If this is a good formulation, next step is to solve  $\nabla F(\mathbf{u}) = \mathbf{0}$ , which in practice would just involve solving a linear system.

3. Solve for forces everywhere on channel boundary:  $\mathcal{M}_{St}\mathbf{f} = \mu\mathbf{u}_b$  on  $\Gamma$  (same as before)
4. Calculate channel velocity on entire domain:  $\mathcal{M}_{St}\mathbf{f} = \mu\mathbf{u}$  in channel (same as before)

# Codes for Segments

- [See Github for current codes: Michaela/Segments vs MRS](#)
- **Infinite\_Channel\_segments\_validation.m**: This code validates the force-to-velocity routine for both the Stokeslet Segment velocity and Source Doublet Segment Velocity on a single segment, comparing outputs to the outputs using the MRS method.
- **segment\_int\_terms.m**: This code computes all the T's and Q's needed for the segment regularizing functions in our solution formula.
- **seg\_reg\_fncs.m**: This code outputs the regularizing function pieces involving  $H_{1,2}$  and  $S_{1,2}$  that we need for the segment solution for a user-specified choice of blob.
- **st\_segments\_forcetovelocity** , **st\_segments\_velocitytoforce**: These implement the force-to-velocity and velocity-to-force routines for the Stokeslet Segment method (Used in Steps 3-4 of Channel Process).
- **sd\_segments\_velocitytogforce.m**, **sd\_segments\_gforcetovelocity.m**: These implement steps 3-4 of the channel process, utilizing source doublets, and treating  $\beta$  as constant on each segment.
- **sd\_segments\_velocitytogforce\_beta\_linear.m**, **sd\_segments\_gforcetovelocity\_beta\_linear.m**: These implement steps 3-4 of the channel process, utilizing source doublets, and treating  $\beta$  as linear on each segment.
- **Infinite\_Channel\_Segment\_Method\_April4.m**: This code implements the channel process using the segment method. It also implements the Stokeslets-only segment method.
- **Infinite\_Channel\_MRS\_Method.m**: This code implements the channel process using MRS



**Background slides for channel  
problem and segment method  
formulation and code**

# 2D Infinite Channel Flow

## Problem Set-up

- Utilizing problem and exact solution from Bernardi et al. (2023)<sup>3</sup>

- Flow driven by pressure difference between interior and exterior

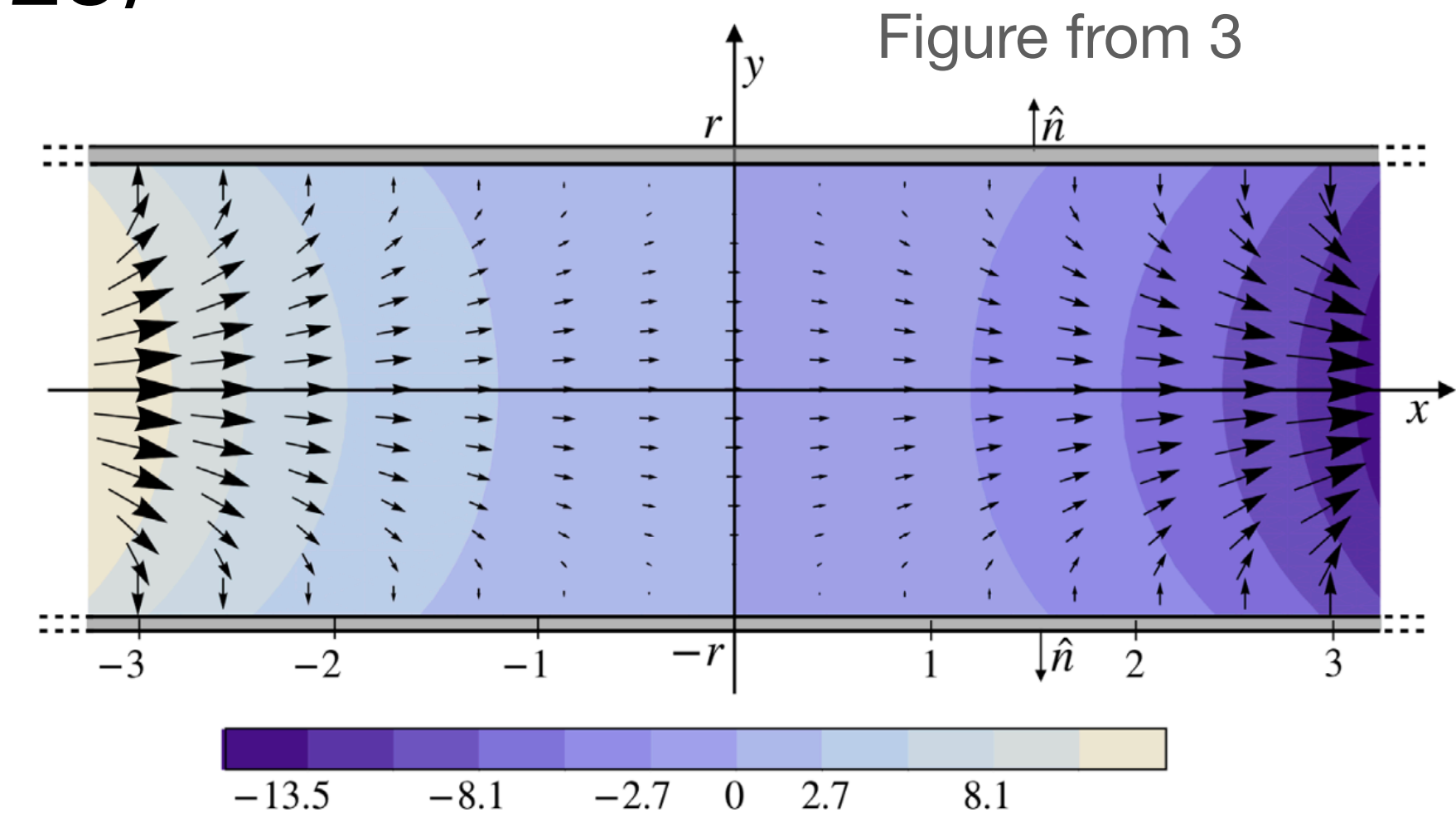
- Boundary condition from no-slip and Darcy's law

$$\mathbf{u} \cdot \hat{\tau} = 0 \text{ and } \mathbf{u} \cdot \hat{\mathbf{n}} = \text{Da } p \text{ on } \Gamma$$

where Da is the Darcy number.

- Solution  $\mathbf{u} = \left( -\frac{C}{\lambda_1} \cosh(\lambda_1 x) g_1'(y), C \sinh(\lambda_1 x) g_1(y) \right)$ ,  $p = C \sinh(\lambda_1 x) \cos(\lambda_1 y)$ ,

$$g_1(y) = \frac{\cos(\lambda_1)}{\sin(\lambda_1)} \left( \text{Da} - \frac{1}{2} \right) \sin(\lambda_1 y) + \frac{1}{2} y \cos(\lambda_1 y) \text{ and } \lambda_1 = \text{root of } \tan^2(\lambda) - \frac{\tan(\lambda)}{\lambda} = 2\text{Da} - 1$$



3. F. Bernardi, S. Chellam, N. Cogan, M. Moore Stokes flow solutions in infinite and semi-infinite porous channels, Studies in Applied Mathematics 151 (1) (2023) 116–140.



# Permeable Channel Solution process

- **Idea:** Given  $\beta(s)$ , find a force distribution,  $\mathbf{g}$ , on  $\Gamma$  with a distribution of SD  $\beta(s)(\mathbf{g}(s) \cdot \hat{\mathbf{n}}(s))$  in  $\Gamma_\beta$  we can use to find missing boundary velocity
- Solving for velocity within the channel is a 4-step process:
  1. Solve for  $\mathbf{g}$  on  $\Gamma$ .  $(\mathcal{M}_{St} + \mathcal{M}_{SD})\mathbf{g} = \mu\mathbf{u}_b$  on  $\Gamma \setminus \Gamma_\beta$ ,  $\mathcal{M}_{St}\mathbf{g} = \mathbf{0}$  on  $\Gamma_\beta$   
(Assume solid boundary velocities are sum of St and SD)  
(Assume permeable boundary velocity (on  $\Gamma_\beta$ ) is entirely due to SD)
  2. Calculate the permeable boundary velocity using the SD distribution:  
 $\mathcal{M}_{SD}\mathbf{g} = \mu\mathbf{u}_b$  on  $\Gamma_\beta$
  3. Solve for forces everywhere on channel boundary:  $\mathcal{M}_{St}\mathbf{f} = \mu\mathbf{u}_b$  on  $\Gamma$
  4. Calculate channel velocity on entire domain:  $\mathcal{M}_{St}\mathbf{f} = \mu\mathbf{u}$  in channel

Regularized  
Source  
Doublet  
Problem

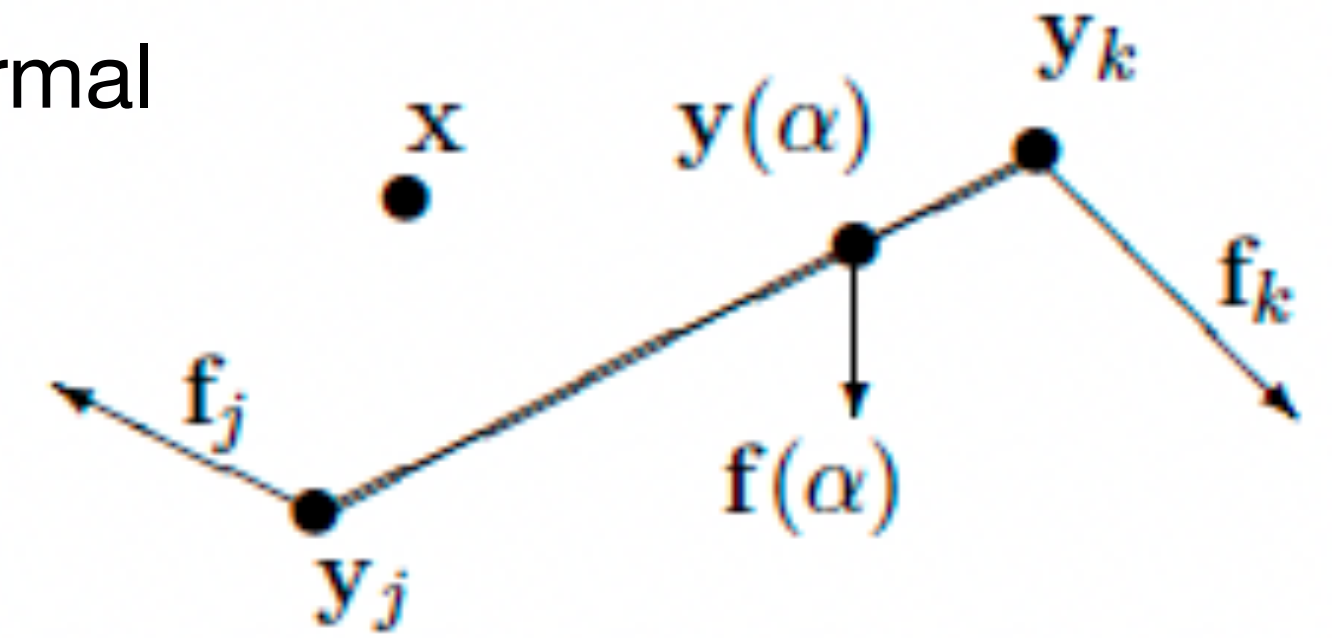
Regularized  
Stokeslet  
Problem

# Segment Method - Quick Summary

(Full formulation notes uploaded into Overleaf!)

On each segment, we have...

- $\mathbf{y}_j$  = segment starting point,  $\mathbf{y}_k$  = segment ending point,  $\mathbf{y}(\alpha) = \mathbf{y}_j + \alpha(\mathbf{y}_k - \mathbf{y}_j)$ ,  $\alpha \in [0,1]$
- $\mathbf{x}_0 = \mathbf{x} - \mathbf{y}_j$ ,  $\mathbf{v} = \mathbf{y}_j - \mathbf{y}_k$  (unnormalized tangent vector),  $\hat{\mathbf{n}}$  = outward unit normal
- $\mathbf{f}(\alpha) = \mathbf{f}_j + \alpha(\mathbf{f}_k - \mathbf{f}_j)$  = linear distribution of force densities on segment
- $H_{1,2}^n = \int_0^1 \alpha^n H_{1,2}(r) d\alpha$ ,  $S_{1,2}^n = \int_0^1 \alpha^n S_{1,2}(r) d\alpha$
- **Stokeslet Segment Velocity** on one segment:  $\mathbf{u}(\mathbf{x}) = \frac{L}{\mu}(A_{St}\mathbf{f}_j + B_{St}\mathbf{f}_k)$ , where  $L$  = segment length
  - $B_{St} = H_1^1 I_{2 \times 2} + H_2^1 X_0 + H_2^2 W + H_2^3 V$ ,  $A_{St} = [H_1^0 I_{2 \times 2} + H_2^0 X_0 + H_2^1 W + H_2^2 V] - B_{St}$
  - $X_0 = \mathbf{x}_0 \mathbf{x}_0^T$ ,  $V = \mathbf{v} \mathbf{v}^T$ ,  $W = \mathbf{x}_0 \mathbf{v}^T + \mathbf{v} \mathbf{x}_0^T$ ,  $N = \hat{\mathbf{n}} \hat{\mathbf{n}}^T$ ,  $Z = \mathbf{v} \mathbf{x}_0^T \hat{\mathbf{n}} \hat{\mathbf{n}}^T$





# Segment Method – Quick Summary Cont'd

- For Source Doublets,  $A(\alpha) = \mathbf{f}(\alpha) \cdot \hat{\mathbf{n}} = \mathbf{f}_j \cdot \hat{\mathbf{n}} + \alpha(\mathbf{f}_k - \mathbf{f}_j) \cdot \hat{\mathbf{n}}$ , where  $\mathbf{f}_j = \beta_0^j \mathbf{g}_j$  ( $\beta_0 = (\beta_j + \beta_k)/2$ )

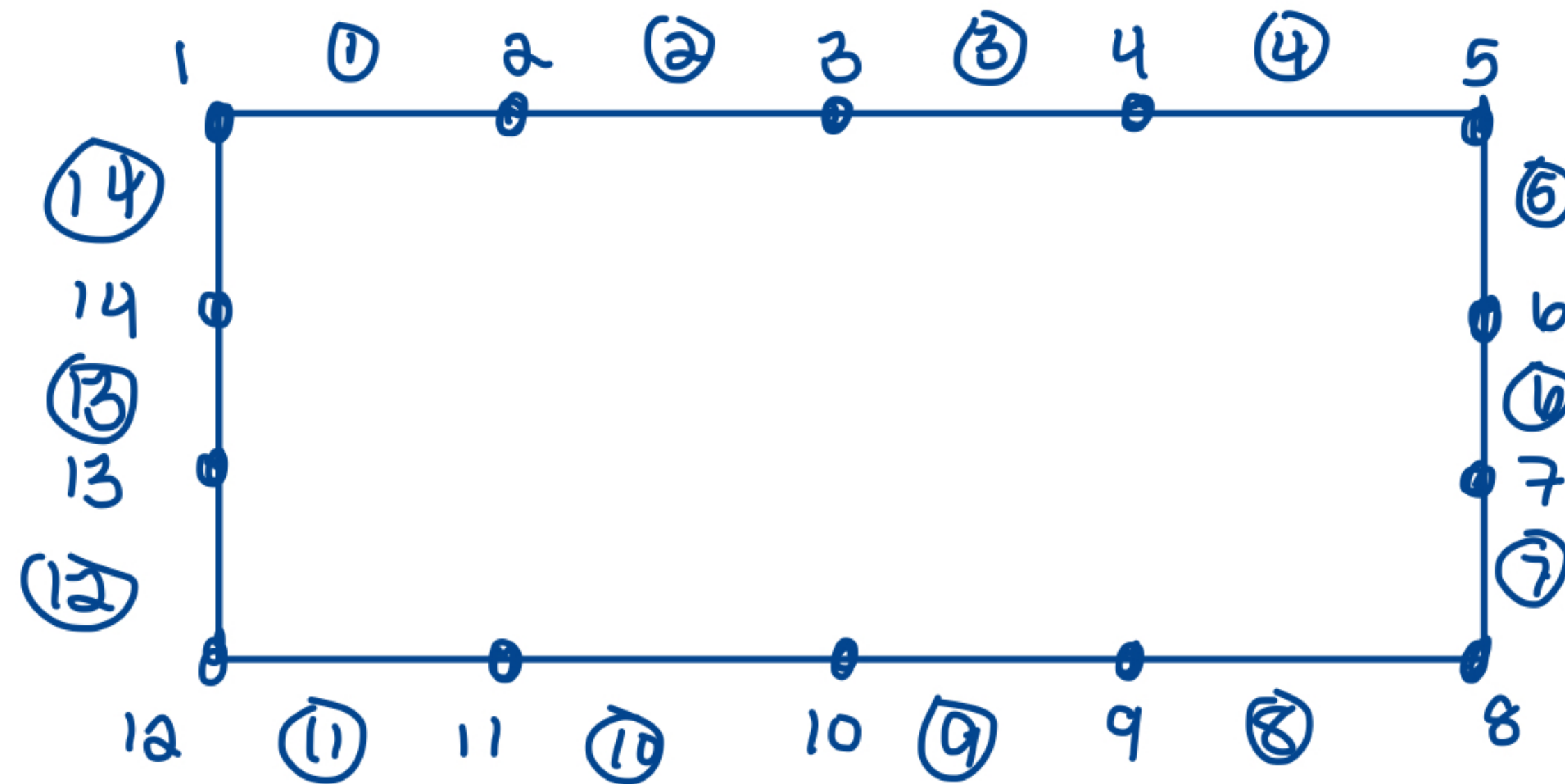
- **Source Doublet velocity** on a single segment:  $\mathbf{u}(\mathbf{x}) = -\frac{L}{\mu}(A_{SD}\mathbf{f}_j + B_{SD}\mathbf{f}_k)$

- $B_{SD} = S_1^1 N + S_2^1 X_0 N + S_2^2 Z$ ,  $A_{SD} = [S_1^0 N + S_2^0 X_0 N + S_2^1 Z] - B_{SD}$

When we have a curve discretized into  $N_{seg}$  line segments,

- Velocity is sum of segment velocities:  $\mathbf{u}(\mathbf{x}) = \frac{1}{\mu} \sum_{i=1}^{N_{seg}} L_i \left( A_{St}^i \mathbf{f}_j^i + B_{St}^i \mathbf{f}_k^i \right)$  and similar for source doublets
- $A_{St}^i, B_{St}^i$  refer to the matrices you get on segment  $i$ , so they depend on  $\mathbf{y}_j$ ,  $\mathbf{x}$ , and the blob.
- $H_{1,2}^n$  and  $S_{1,2}^n$  will vary by blob, but will consist of terms in the form  $T_{n,q} = \int_0^1 \frac{\alpha^n}{R^q} d\alpha$ ,  $Q_n = \int_0^1 \alpha^n \log(R) d\alpha$ , where  $R^2 = r^2 + \epsilon^2$ , and can be calculated exactly.

# Some info on how the codes work



$y = [y_{top}(left-right);$   
 $y_{right}(top-bot);$   
 $y_{bot}(right-left);$   
 $y_{left}(bot-top)]$

segment

$segData^T =$

|                 |           |           |           |           |       |       |       |           |              |              |              |          |          |          |
|-----------------|-----------|-----------|-----------|-----------|-------|-------|-------|-----------|--------------|--------------|--------------|----------|----------|----------|
| idx_start       | 1         | 2         | 3         | 4         | 5     | 6     | 7     | 8         | 9            | 10           | 11           | 12       | 13       | 14       |
| idx_end         | 2         | 3         | 4         | 5         | 6     | 7     | 8     | 9         | 10           | 11           | 12           | 13       | 14       | 1        |
| ep              | $e_1$     | $e_2$     | $e_3$     | $e_4$     | $e_5$ | $e_6$ | $e_7$ | $e_8$     | $e_9$        | $e_{10}$     | $e_{11}$     | $e_{12}$ | $e_{13}$ | $e_{14}$ |
| $\beta_{start}$ | 0         | $\beta_2$ | $\beta_3$ | $\beta_4$ | 0     | 0     | 0     | 0         | $\beta_9$    | $\beta_{10}$ | $\beta_{11}$ | 0        | 0        | 0        |
| $\beta_{end}$   | $\beta_2$ | $\beta_3$ | $\beta_4$ | 0         | 0     | 0     | 0     | $\beta_9$ | $\beta_{10}$ | $\beta_{11}$ | 0            | 0        | 0        | 0        |

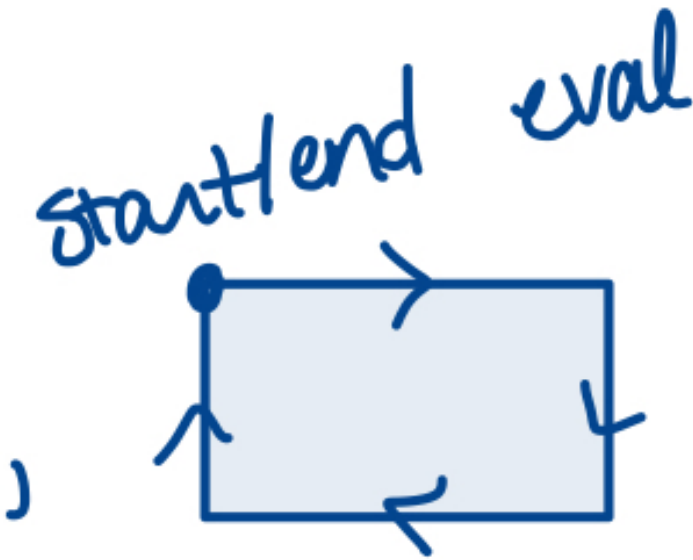


# Closed boundary

$$u(x) = \frac{1}{N} \sum_{i=1}^{N_{\text{seg}}} (A_{st}^i f_f^i + B_{st}^i f_k^i)$$

$\uparrow$  start       $\nwarrow$  end

When  $N_{\text{seg}} = N_{\text{nodes}}$  (closed boundary),



$$A_{11}^i = A_{11}(y_i, y_j)$$

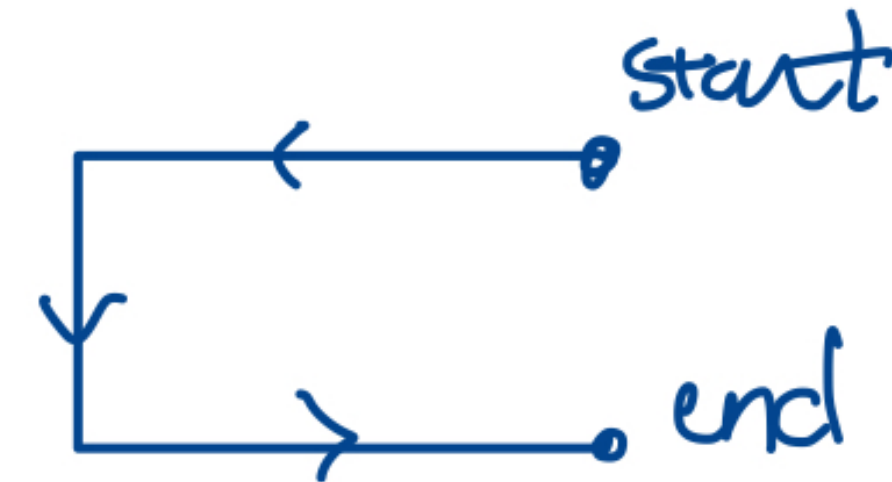
$\nwarrow$  segment

$$\begin{bmatrix} u(y_1) \\ u(y_2) \\ \vdots \\ u(y_N) \\ v(y_1) \\ v(y_2) \\ \vdots \\ v(y_N) \end{bmatrix} = \begin{bmatrix} A_{11}^{11} + B_{11}^{1,N} & A_{11}^{12} + B_{11}^{11} & A_{11}^{13} + B_{11}^{12} & \dots & A_{11}^{1,N} + B_{11}^{1,N-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A_{11}^{N,1} + B_{11}^{N,N} & A_{11}^{N,2} + B_{11}^{N,1} & A_{11}^{N,3} + B_{11}^{N,2} & \dots & A_{11}^{N,N} + B_{11}^{N,N-1} \\ \hline A_{21}^{11} + B_{21}^{1,N} & A_{21}^{12} + B_{21}^{11} & A_{21}^{13} + B_{21}^{12} & \dots & A_{21}^{1,N} + B_{21}^{1,N-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A_{21}^{N,1} + B_{21}^{N,N} & A_{21}^{N,2} + B_{21}^{N,1} & A_{21}^{N,3} + B_{21}^{N,2} & \dots & A_{21}^{N,N} + B_{21}^{N,N-1} \\ \hline A_{22}^{11} + B_{22}^{1,N} & A_{22}^{12} + B_{22}^{11} & A_{22}^{13} + B_{22}^{12} & \dots & A_{22}^{1,N} + B_{22}^{1,N-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A_{22}^{N,1} + B_{22}^{N,N} & A_{22}^{N,2} + B_{22}^{N,1} & A_{22}^{N,3} + B_{22}^{N,2} & \dots & A_{22}^{N,N} + B_{22}^{N,N-1} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_{N-1} \\ f_N \\ g_1 \\ g_2 \\ \vdots \\ g_{N-1} \\ g_N \end{bmatrix}$$



# Open boundary

When  $N_{seg} = N_{nodes} - 1$  (not closed)



$$\begin{bmatrix} u(y_1) \\ u(y_2) \\ \vdots \\ u(y_N) \\ v(y_1) \\ v(y_2) \\ \vdots \\ v(y_N) \end{bmatrix} = \begin{bmatrix} A_{11}^{11} & A_{11}^{12} + B_{11}^{11} & A_{11}^{13} + B_{11}^{12} & \dots & A_{11}^{1,N} + B_{11}^{1,N-1} & B_{11}^{1,N} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{11}^{N1} & A_{11}^{N2} + B_{11}^{N1} & A_{11}^{N3} + B_{11}^{N2} & \dots & A_{11}^{N,N} + B_{11}^{N,N-1} & B_{11}^{N,N} \\ \hline A_{12}^{11} & A_{12}^{12} + B_{12}^{11} & A_{12}^{13} + B_{12}^{12} & \dots & A_{12}^{1,N} + B_{12}^{1,N-1} & B_{12}^{1,N} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{12}^{N1} & A_{12}^{N2} + B_{12}^{N1} & A_{12}^{N3} + B_{12}^{N2} & \dots & A_{12}^{N,N} + B_{12}^{N,N-1} & B_{12}^{N,N} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_{N-1} \\ f_N \\ g_1 \\ g_2 \\ \vdots \\ g_{N-1} \\ g_N \end{bmatrix}$$

$$N = N_{nodes} = N_{seg} + 1$$

$$u(x) = \frac{1}{\mu} \sum_{i=1}^{N_{seg}} (A_{st}^i f_g^i + B_{st}^i f_k^i)$$